

Reproduce-then-Extend — Pinyon Jay Habitat Selection (Ecology)

(Day 2)

Intro to Machine Learning · Bruce A. Desmarais

July 03, 2026

Published study. Daw, S. K. et al. (2022). “Behavior-specific occurrence patterns of Pinyon Jays.” *PLoS ONE* (doi:10.1371/journal.pone.0237621; S1 Dataset, open). A **logistic resource-selection function (RSF)** — habitat *use* points (1) vs *available* points (0) on environmental predictors. Use is the minority class, so out-of-sample skill is scored by **AUC-PR** (area under the precision-recall curve).

1 Background

Pinyon Jays are a **declining bird** of the southwestern United States, and managers need to know **which habitat they use** in order to conserve it. Daw and colleagues fit a **resource-selection function** – a logistic regression comparing locations the jays *used* to *available* background locations – to identify the environmental conditions associated with jay occurrence. The **primary conclusion** is that jays select particular pinyon-juniper conditions – more open stands, sites near habitat edges, and areas with downed woody debris – which the regression translates into a picture of suitable habitat for management.

2 Data and codebook

Unit of analysis: a sampled ground location – an observed jay-**use** point vs an **available** (background) point; $n = 466$ (124 use).

Variable	Definition
use	1 = observed Pinyon Jay use location, 0 = available location (outcome)
Elevation..feet.	elevation (feet)
Slope..percent.	ground slope (%)
Stand_Age..years.	forest stand age (years)
Stand_Density_Index	stand density index (tree density / competition)
Canopy_Cover..percent.	overstory canopy cover (%)
Tree_Cover..percent.	tree cover (%)
Shrub_Cover..percent.	shrub cover (%)
Forb_Cover..percent.	forb (herbaceous) cover (%)
Grass_Cover..percent.	grass cover (%)
Woody_Debris..category.1.-.3.	counts of downed woody debris by size class (1-3)
Distance.to.Road..km.	distance to the nearest road (km)
DisturbanceYes	1 if any recent disturbance was recorded at the site
Distance.to.Edge..m.	distance to the nearest habitat edge (m)

3 Descriptive results

```
d <- read.csv("data/jays_clean.csv")

# Predictors: keep the interpretable environmental variables and drop the sparse one-hot
# habitat/disturbance-type dummies (unidentified in the logit).
preds <- setdiff(names(d), "use")
preds <- preds[sapply(d[preds], function(x) length(unique(x)) > 1)]
preds <- preds[!grepl("Habitat_Type|Disturbance_Type", preds)]

# Descriptive statistics for the outcome and the first several predictors
stats <- t(sapply(d[c("use", head(preds, 8))],
  function(x) c(mean = mean(x), sd = sd(x), min = min(x), max = max(x))))
round(stats, 2)
```

##		mean	sd	min	max
##	use	0.27	0.44	0	1.00
##	Elevation..feet.	6809.96	686.14	4430	8700.00
##	Slope..percent.	28.42	17.64	0	91.00
##	Stand_Age..years.	143.20	81.17	0	548.00
##	Stand_Density_Index	154.74	97.50	0	542.04
##	Canopy_Cover..percent.	27.92	15.98	0	74.00
##	Tree_Cover..percent.	19.05	13.42	0	59.75
##	Shrub_Cover..percent.	6.07	5.98	0	37.25
##	Forb_Cover..percent.	1.57	2.60	0	26.00

```
clean <- function(v) gsub("[._]+", " ", v) # tidy the mangled column names for labels

par(mfrow = c(1, 2))
# (left) class balance of the used-vs-available outcome
par(mar = c(4, 4, 2.5, 1))
barplot(table(d$use), names.arg = c("available", "used"),
  col = c("grey75", "#1f78b4"), main = "Outcome: use", ylab = "count")

# (right) standardized mean difference (used - available) for each predictor
smd <- sapply(d[preds], function(x) (mean(x[d$use == 1]) - mean(x[d$use == 0])) / sd(x))
smd <- tail(smd[order(abs(smd))], 10) # the 10 largest-magnitude differences
par(mar = c(4, 8.5, 2.5, 1))
barplot(smd, horiz = TRUE, las = 1, names.arg = clean(names(smd)), cex.names = 0.75,
  col = ifelse(smd > 0, "#1f78b4", "#e31a1c"), main = "Used vs available",
  xlab = "std. mean difference")
abline(v = 0, col = "grey50")
```

4 Reproduce the published RSF regression

The published model is a logistic regression of use vs availability on the environmental predictors.

```
rsf <- glm(reformulate(preds, "use"), data = d, family = binomial)
round(summary(rsf)$coefficients, 3)
```

##		Estimate	Std. Error	z value	Pr(> z)
##	(Intercept)	6.735	2.827	2.383	0.017
##	Elevation..feet.	0.000	0.000	-1.080	0.280
##	Slope..percent.	-0.067	0.015	-4.495	0.000

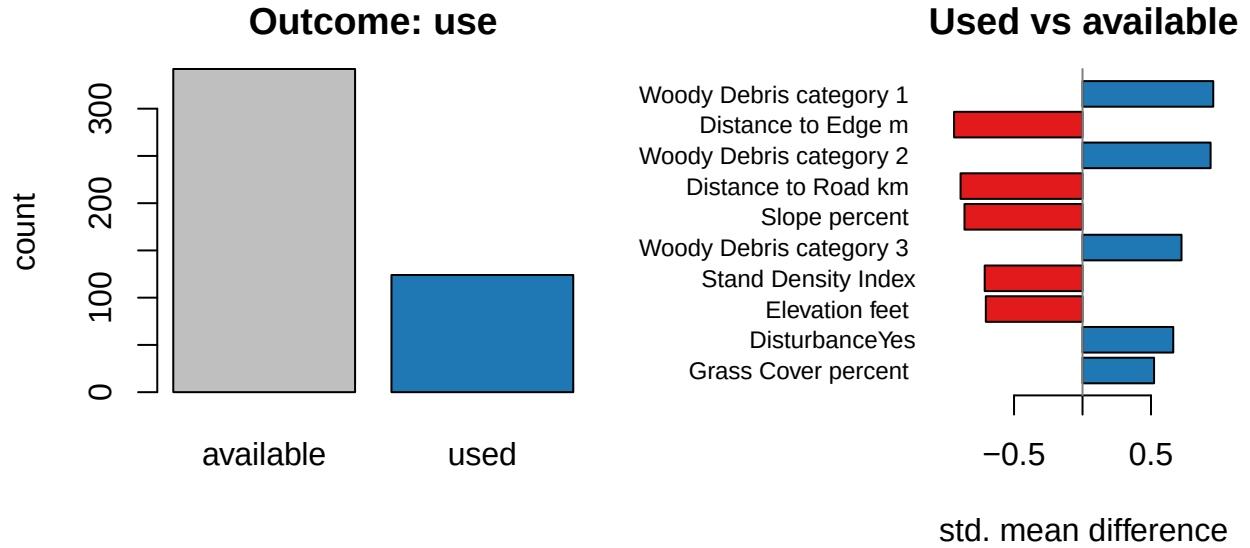


Figure 1: Outcome balance (left) and the environmental variables that differ most between used and available points (right).

```
## Stand_Age..years.      0.001      0.003      0.322      0.748
## Stand_Density_Index    -0.024      0.005     -4.699      0.000
## Canopy_Cover..percent.  0.034      0.023      1.500      0.134
## Tree_Cover..percent.   0.080      0.026      3.108      0.002
## Shrub_Cover..percent.  -0.015      0.035     -0.435      0.664
## Forb_Cover..percent.   -0.056      0.084     -0.668      0.504
## Grass_Cover..percent.  -0.039      0.027     -1.464      0.143
## Woody_Debris..category.1.  0.029      0.014      2.132      0.033
## Woody_Debris..category.2.  0.197      0.076      2.597      0.009
## Woody_Debris..category.3.  0.005      0.105      0.047      0.962
## Distance.to.Road..km.    -0.452      0.134     -3.383      0.001
## DisturbanceYes          2.269      0.559      4.055      0.000
## Distance.to.Edge..m.    -0.001      0.000     -4.763      0.000
```

Confirmation against the published study. The signed coefficients recover the reported habitat-selection pattern — jays select more **open** stands (negative on `Stand_Density_Index` and `Slope..percent.`) and sites **closer to edges** (negative on `Distance.to.Edge..m.`). This logistic RSF is our regression baseline.

5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner's hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$use
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
  ↪ 0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
        sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]
```

```

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(use ~ ., data = d[tr, c("use", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
  ↪ = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
best_rf <- NULL
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
  ↪ 2)))) {
  rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
  ↪ TRUE)
  sc <- aucpr(rf$predictions[, "1"], y[tr]) # OOB predictions = a training-only
  ↪ CV estimate
  if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
  ↪ probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
  ↪ stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
  ↪ eta,
                                subsample = 0.9, colsample_bytree = 0.8, eval_metric =
                                ↪ "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
  ↪ nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
  ↪ = nr, score = score)
}
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
  ↪ eta = best$eta,
                                subsample = 0.9, colsample_bytree = 0.8),
                  data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)

##          Logit   ElasticNet RandomForest      XGBoost
##          0.953      0.970      0.969      0.970

cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
  ↪ AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))

```

Best ML gain over the logistic RSF on the held-out test set: +0.018 AUC-PR

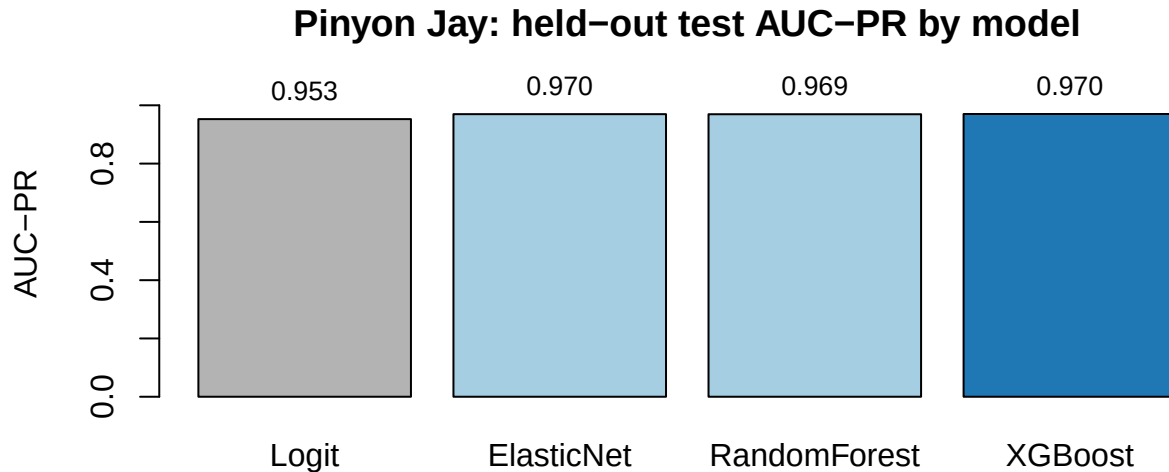


Figure 2: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

On the held-out test set, random forest and XGBoost lift AUC-PR over the logistic RSF — the nonlinear, interaction-rich structure of habitat selection that a linear-in-the-logit model cannot capture.

6 Interpreting the model: importance, SHAP, and partial dependence

We open the fitted learners three ways: **permutation importance**, **SHAP values** (exact TreeSHAP — the signed contribution of each variable to each prediction), and **partial dependence** (the shape of an effect).

```
# Random forest (permutation importance + partial dependence) on all the data
rf <- ranger(x = X, y = factor(y), num.trees = 800, probability = TRUE, importance =
  ↪ "permutation")

# XGBoost fit on all the data, then exact TreeSHAP contributions per observation
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,
  subsample = 0.9, colsample_bytree = 0.8),
  data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE) # last column is the model bias
shap <- shap[, colnames(X)]                # drop the BIAS column
```

Interpreting the relationships. The SHAP summary reads directly as habitat selection: use rises as **stand-density index falls** (open blue points sit at positive SHAP, dense red at negative) and **closer to edges**, with **woody debris** present. In plain terms, pinyon jays select **open pinyon-juniper woodland near edges with downed wood** — and avoid dense, closed stands.

Both curves show a strongly **nonlinear** effect — the contribution turns over across the range rather than following one logit slope, which (with the interactions SHAP captures locally) is why the learners beat the linear RSF.

7 Takeaway

Birds select habitat at nonlinear optima and via interactions — where the linear RSF struggles and trees excel — and SHAP lets us read those relationships (open stands, edges, woody debris) directly off the fitted model. The first of two ecology cases showing ML both predicts better *and* interprets cleanly.

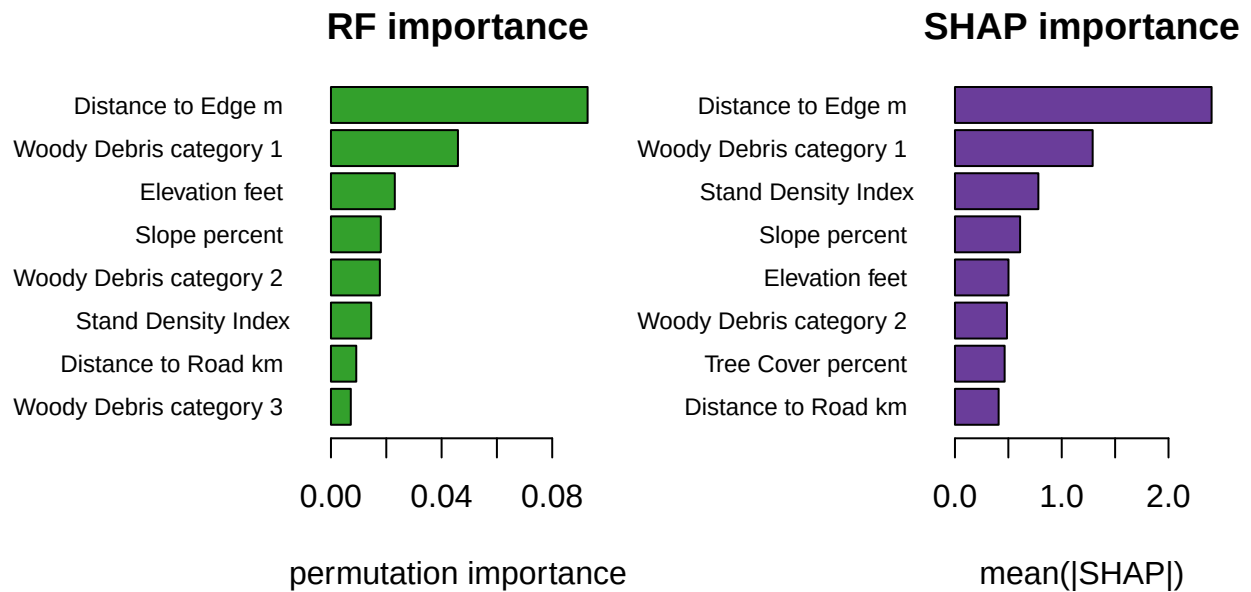


Figure 3: Permutation importance (left) and mean-absolute-SHAP importance (right). The two rankings agree.

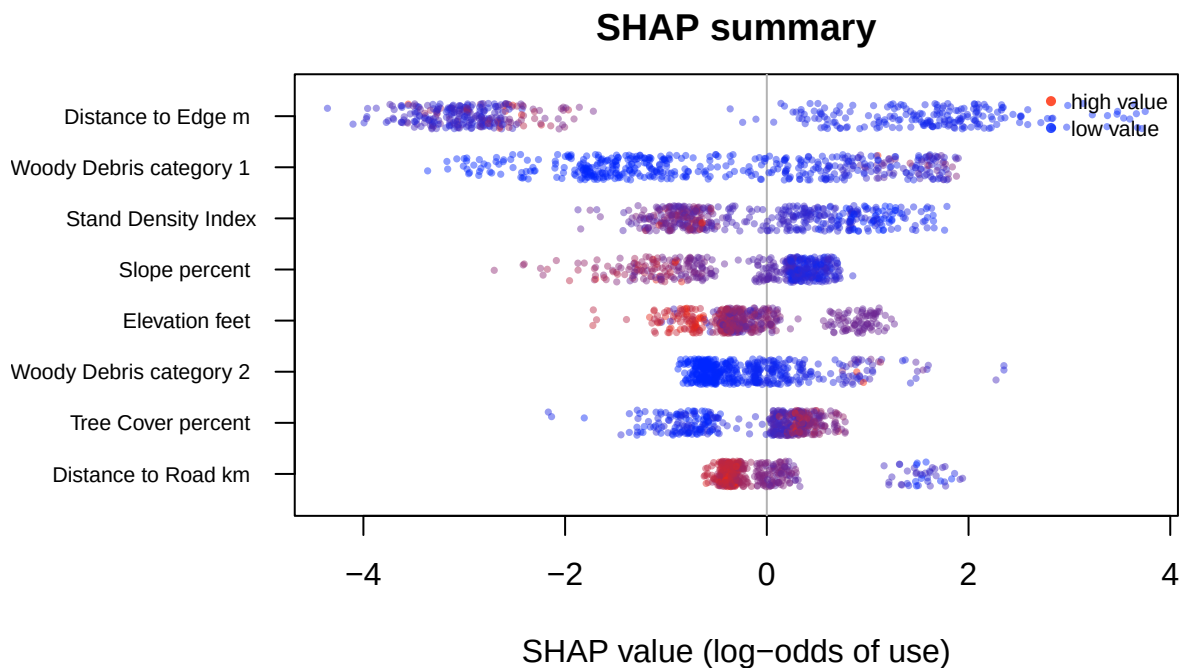


Figure 4: SHAP summary. Each point is one location; x = its SHAP contribution to the log-odds of use; colour = the variable's value (blue low, red high).

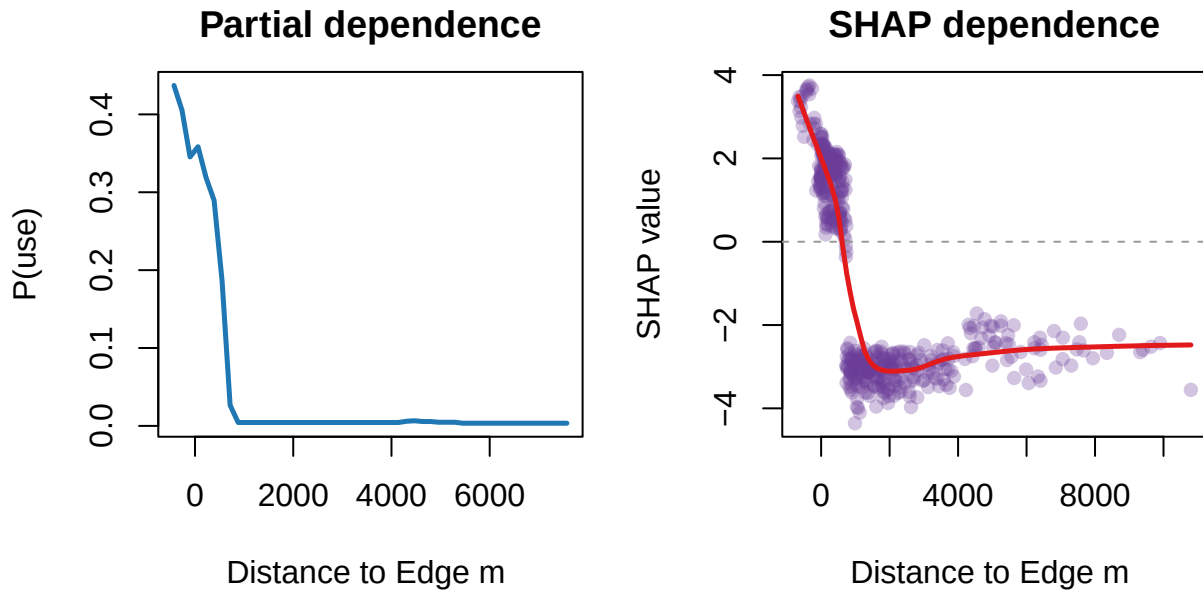


Figure 5: Partial dependence (left) and SHAP dependence (right) for the most influential variable.

8 Recommended exercises

1. Vary the random-forest hyperparameters (number of trees, features per split) and report the effect on AUC-PR.
2. Generate a partial-dependence plot for a second habitat variable and describe the nonlinearity.
3. Compare RF vs XGBoost AUC-PR on the same folds; which wins, and by how much?